

Name:

University ID:

Please, answer the theoretical questions in detail, using only the provided sheets. The exam is **open-book**, and you can use **no electronic devices** except for a portable calculator. All questions and exercises will be corrected extensively; you are encouraged to expand beyond the strict topic of the question (without losing track of the target), like in an oral exam. You are also encouraged to draw sketches for concepts that are visual in nature.

The exam duration is **1h30m**.

This written exam is valid for 32 points. The lab assignments provide up to extra 4 points. Evaluations beyond 30/36 will be graded with 30 with honours.

Question 1

- A) Given the function $f(x, y) = (x - 2y)^4 + (yx + 3x)^2$, evaluate the gradient at $(x, y) = (1, -2)$.
- B) Describe how a Convolution can be transformed into a Matrix Multiplication, which type of overhead this process generates and why it is usually more efficient to execute MatMul-like convolutions than pure convolutions in digital processors (e.g. specialized CPUs, GPUs, Tensor cores).
- C) Write a formula to estimate the number of MAC operations for a pointwise convolution with N input channels (256x64 size) to produce 48 output channels
- D) How many weights do you need to store a depth-wise + pointwise convolution with 3x3 filters, 16x16 sized spatial dimension, 32 input channels, and 256 output channels? Assume padding with zeros.

A)

$$\begin{aligned}\frac{df}{dx} &= 4(x - 2y)^3 + 2x(y + 3)^2, \\ \frac{df}{dy} &= -8(x - 2y)^3 + 2x^2(y + 3)\end{aligned}$$

Then,

$$\frac{df}{dx}(1, -2) = 4(1 + 4)^3 + 2(-2 + 3)^2 = 4 * (5^3) + 2 = 502$$

$$\frac{df}{dy}(1, -2) = -8(1 + 4)^3 + 2(-2 + 1) = -8 * (5^3) + 2 = -998$$

Name:

University ID:

B)

Convolution -> matrix multiplication using Im2Col. More efficient because of regular data access + possible reuse.

C)

$$\text{MAC_PW} = \text{Cin} * \text{Ho} * \text{Wo} * \text{Cout} = N * 256 * 64 * 48 = N * 786.432$$

D)

$$\text{W_DW} = \text{Cin} * \text{Hk} * \text{Wk} = 32 * 3 * 3 = 288$$

$$\text{W_PW} = \text{Cin} * \text{Cout} = 32 * 256 = 8129$$

TOTAL = 8480 weights

Name:

University ID:

Question 2

A parallel cluster of RISC-V processors is executing a matrix multiplication $A*B$ of matrices A (32×64) and B (64×32). Assume that the computation is parallelized assigning tiles of size 8×8 of the output matrix C to processors, following a row-major tile order. The processor cluster has 8 processors. Elements of A , B , C are 8bit FP numbers

- **A) Compute the L1 minimal shared memory size that is needed to store the complete matrices, and the total number of MACs needed to complete the matrix multiplication.**
- **B) Compute the number of tiles to be assigned to each core, trying to minimize the workload unbalance. If the number is not identical for all the cores, give min and max.**
- **C) Compute the speedup with respect to sequential computation. Assume 21% of stall cycles due to banking conflicts.**

A)

$$\text{MEM_L1} = \text{MEM_A} + \text{MEM_B} + \text{MEM_C} = 32 \times 64 + 64 \times 32 + 32 \times 32 \text{ B} = 5.120 \text{ B} = 40.960 \text{ b}$$

$$\text{MAC} = 32 \times 64 \times 32 = 65.536$$

B)

Output size: $32 \times 32 \rightarrow (32/8) \times (32/8) = 4 \times 4 = 16$ tiles $\rightarrow$ fully balanced workload
Every core takes 2 tiles in parallel.

C)

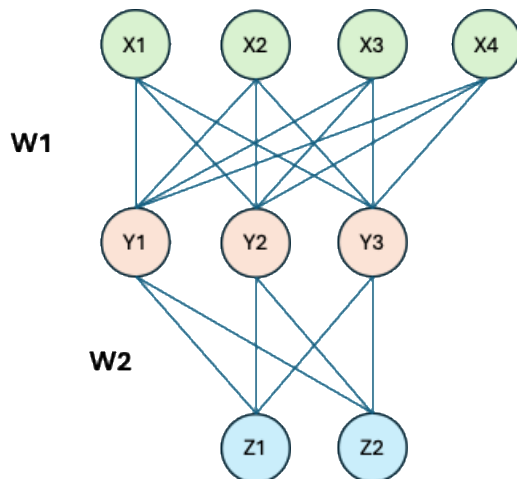
$$\text{Speed up} = (\text{single core})/(\text{parallel}) = 65536 / (65536/8 * 1.21) = 65536 / 9913 = 6.6 \times$$

Name:

University ID:

Question 3

Consider the MLP in Figure. The first layer is described by the Matrix $W1$, assuming no bias values; the second layer is described by the Matrix $W2$. The activation function for both layers is the ReLu.



$W1$

1.1	0	-1	1
0	1	1	1.1
0.9	-2.1	6.1	0

$W2$

0	0.9	-0.8
-1	0	1

- A)** Assume at the input you have the sample $S1$ described by the column vector $(3, 2, 1, 0.7)$, the sample $S2$ described by the column vector $(-1.5, 3, 1.8, 2.1)$, and the sample $S3$ described by column vector $(0, 0, 1, 0)$ compute the class scores $(Z1, Z2)$ of the MLP associated to $S1, S2, S3$.
- B)** Consider the multi-class SVM loss function. Compute the loss value for $S1$ and $S2$, assuming that $S1$ and $S2$ have label $C2$, and $S3$ has label $C1$.
- C)** Quantize the weights of $W2$ in unit8. First compute the scale factor (S) and the zero point (Z), then derive the quantized matrix $QW2$.
- D)** Re-compute the multi-class SVM loss function as at point B), but using $QW2$ instead of $W2$. Assume same scale and zero point of $W2$. How does the loss change? Does this affect the final classification result? Comment deeply on the results obtained.

A)

$S1$:

$$Y1 = W1 * S1 = (3, 3.77, 4.6)$$

$$Y1_relu = \text{relu}(Y1) = (3, 3.77, 4.6)$$

$$Y2 = W2 * S2 = (-0.287, 1.6)$$

$$\text{Out} = \text{relu}(Y2) = (0, 1.6)$$

$S2$:

Name:

University ID:

$$Y1 = W1 * S1 = (-1.35, 7.11, 3.33)$$

$$Y1_relu = \text{relu}(Y1) = (0, 7.11, 3.33)$$

$$Y2 = W2 * S2 = (3.735, 3.33)$$

$$\text{Out} = \text{relu}(Y2) = (3.735, 3.33)$$

S3:

$$Y1 = W1 * S1 = (-1, 1, 6.1)$$

$$Y1_relu = \text{relu}(Y1) = (0, 1, 6.1)$$

$$Y2 = W2 * S2 = (-3.98, 6.1)$$

$$\text{Out} = \text{relu}(Y2) = (0, 6.1)$$

B)

$$S1: C2 \rightarrow \text{Loss} = \max(0, 0 - 1.6 + 1) = 0$$

$$S2: C2 \rightarrow \text{Loss} = \max(0, 3.33 - 3.735 + 1) = 0.595$$

$$S3: C1 \rightarrow \text{Loss} = \max(0, 6.1 - 0 + 1) = 7.1$$

C)

INPUT MATRIX:

[[0, 0.9, -0.8]

[-1, 0, 1]]

$$s = 0.0078$$

$$z = 0$$

QUANTIZED MATRIX:

[[128, 243, 26]

[0, 128, 255]]

D)

STEP 1: quantize activations -> in this case, we suppose same scale and zero point of the weights of layer 2.

S1:

$$Y1_relu = (3, 3.77, 4.6)$$

Name:

University ID:

$Y1_{\text{quant}} = (255, 255, 255)$
 $Y2_{\text{quant}} = (255, 255)$
 $\text{Out} = \text{relu}(Y2_{\text{quant}}) = (255, 255)$
 $\text{Loss} = \max(0, 255 - 255 + 1) = 1$

S2:

$Y1_{\text{relu}} = (0, 7.11, 3.33)$
 $Y1_{\text{quant}} = (128, 255, 255)$
 $Y2_{\text{quant}} = (255, 255)$
 $\text{Out} = \text{relu}(Y2_{\text{quant}}) = (255, 255)$
 $\text{Loss} = \max(0, 255 - 255 + 1) = 1$

S3:

$Y1_{\text{relu}} = (0, 1, 6.1)$
 $Y1_{\text{quant}} = (128, 255, 255)$
 $Y2_{\text{quant}} = (255, 255)$
 $\text{Out} = \text{relu}(Y2_{\text{quant}}) = (255, 255)$
 $\text{Loss} = \max(0, 255 - 255 + 1) = 1$

This quantization with same scale and zero point saturates the values, therefore preventing correct predictions (mixed precision between floating point and integers + hard saturation).